Lecture XII - Metro Passenger Flow Control

Applied Optimization with Julia

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Introduction

FIFA World Cup 2022 in Qatar

Learning Objectives

After this lecture, you will be able to:

- **Model** time-expanded passenger flows with precomputed delays
- **Translate** a safety goal into inflow control constraints
- **Explain** how the set $\mathcal{R}_{e,t}$ connects periods and minutes
- **Embed** an optimization model into a rolling-horizon heuristic

Public transport

- FIFA World Cup 2022 took place in a **small region**
- The capacity of the metro system was **finite**
- More than **1 million tourists** were expected
- Metro usage was **free for all ticket holders**
- Transport methods were expected to be **overloaded**

Question: What could become an issue?

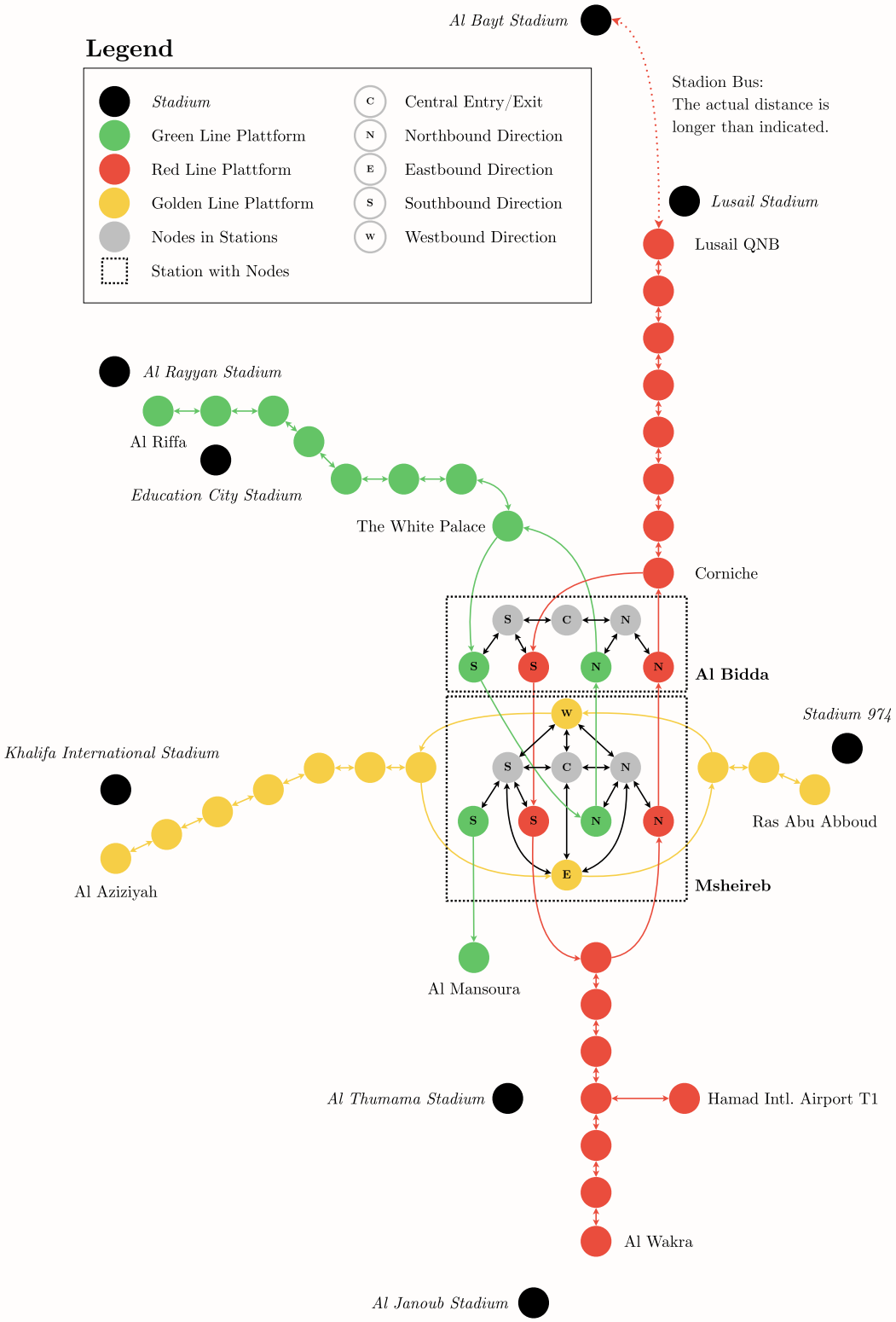
Crowd Disasters

Potential locations

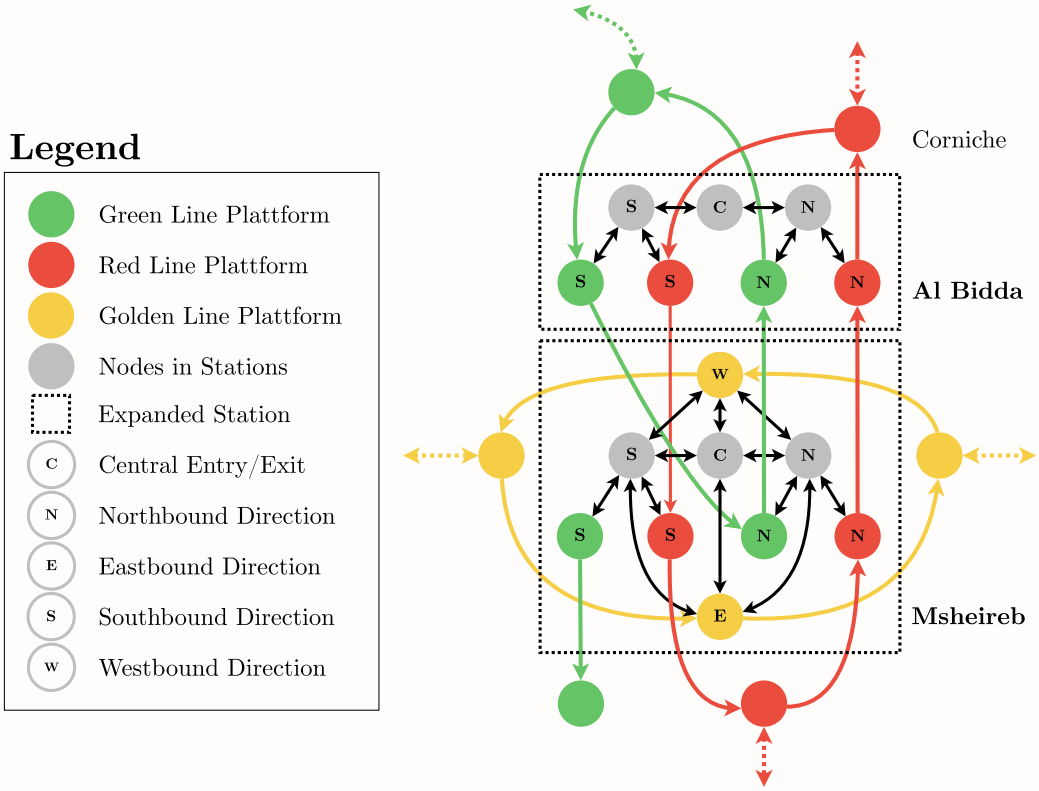
Question: Where could crowd disasters happen?

- At event venues and in waiting areas
- At intersections of multiple pedestrian flows
- In narrow passages and entrances
- On crowded metro platforms and transfer stations
- At ticket/turnstile bottlenecks and emergency exits

Metrosystem of Doha



Closer look at the Metro System



Main Bottleneck

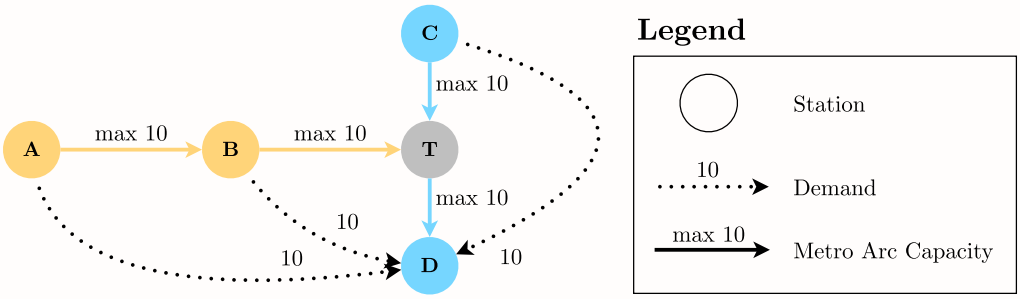
Question: What could be a potential bottleneck?

- Red Line has twice the capacity of Green and Gold Line
- People use the metro to get to the event venues

Question: What could be the issue?

- Imagine a match at a stadium along the Gold line
- People from Red and Green line will try to get there
- This will cause a **massive crowd at transfer stations**

Visualization



- 10 people want to start per minute at **each station**
- Everybody wants to get to **station D**

Capacities inside the stations

- Capacities **inside the stations** are limited as well
- These include the escalators, stairs, and elevators
- But also the platforms and the ticket gates

> ⚠ **Warning**
> This can lead to **overcrowding and potential crowd disasters!**

General Issues

Question: What could also be an issue?

- Often unknown **how many people** exactly will participate
- Gathering data is **extremely important and difficult**
- Crowd behavior can be **unpredictable and dynamic**
- Weather conditions may affect **transportation preferences**
- Cultural factors influence **crowd movement patterns**
- Emergency situations require **flexible contingency plans**

Let's take a

closer look at the

problem structure!

Problem Structure

Objective

Question: What could be the objectives of the authorities?

- A **safe and successful event** as host country
- Good **publicity and a positive recognition** worldwide
- **Satisfied visitors** that enjoyed their time

> **Note: We can model none of the above directly!**
> But we can assist in the estimation of transport demands and the design of operational plans to ensure public **safety** and a **smooth transportation** through the city.

Underlying Problem

Question: Can we just start modeling?

- First, we need to understand the **movement patterns**
- Event is **unprecedented**, movement patterns are unknown
- Multiple **concurrent events** affect flow patterns

> **Tip: We need to estimate the data first!**
> This is a **huge challenge**, but we can use **simulation** to estimate the data.

Simulation

- Based on **publicly available data** from the area
- Simulates **all individuals** participating at the event
- Includes **all transportation infrastructure**
- **Individual mode choice** based on a choice model

> **Tip: Julia is a great tool for this!**
> The simulation was built in Julia. With 1,000,000 individuals walking, using cars, buses, and the metro, simulating a day took **less than 5 minutes**.

Results of the Simulation



- Detailed movement patterns throughout the region
- Potential sections at risk in the transport infrastructure
- Potential capacity overloads at event locations

But it is still

based on a lot

of assumptions!

Main risk: Metro

Idea

Question: What can we do to prevent crowd disasters?

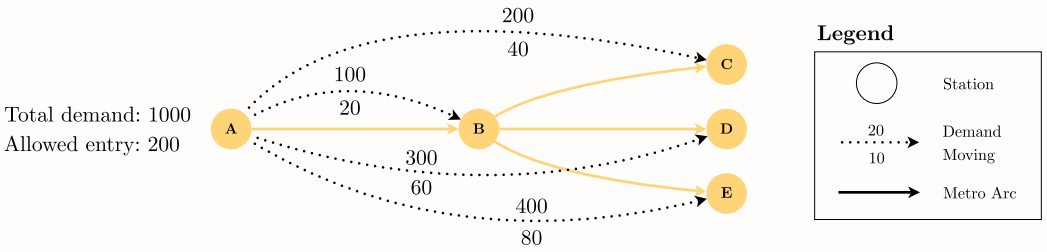
- Regulate the **inflow** at each individual station

- Ensure utilized capacity is **always within bounds**

Question: What could we try to model?

- **Minimize the queues** outside of the metro stations
- Based on the **allowed inflow** and origin-destination data
- Adhering to the **capacity constraints**

Difficulty



- Patterns have different origin-destination pairs
- Regulating the inflow does not affect the destination

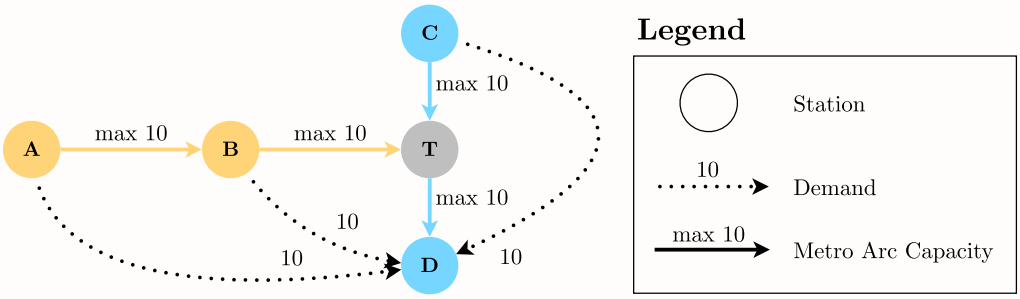
Your Turn

Play: https://beyondsimulations.github.io/Applied-Optimization/games/metro.html

Model Formulation

Graph Sets?

Question: What could be the sets for the graph?



Graph Sets?

- $\mathcal{G}$ - Connected digraph of the metro network $\mathcal{G} = (\mathcal{O}, \mathcal{E})$
- $\mathcal{O}$ - Set of metro stations, indexed by o

- $\mathcal{E}$ - Set of directed arcs between connected stations

> **Note: These are the easier sets.**
> These simply help us to represent the entire metro system as a graph with different nodes $\mathcal{O}$ and arcs $\mathcal{E}$ between connected stations.

Time Sets?

Question: What could be the time sets?

- $\mathcal{T}$ - Set of minutes in the time horizon, indexed by t
- $\mathcal{P}$ - Set of periods in the observed time horizon, where $p \in \{1, 2, ..., n\}$

> **Note: Number of periods**
> We further define n as the **number of periods in observed time horizon.**

Periods

Question: Why do we add periods here?

- Staff needs **clear**, consistent instructions
- Frequent changes in flow **increase risk of errors**
- **Easier** to manage and communicate for planners

> **Note: Period Length**
> We define the period length as the length of the period in minutes measured as m minutes, which is the **same for all periods**.

Mapping Minutes to Periods

- We can define an **additional set** with their relation
- It specifies the relation of **periods p to minutes t.**

$$I_p = \{t \in \mathcal{T} | (p-1) \times m + 1 \leq t \leq p \times m\} \quad \forall p \in \mathcal{P}$$

Question: Can anybody explain it?

- I_p is the set of minutes t that belong to period p
- Minutes are **not overlapping**, but they are **continuous**

The Same Set in Julia

With periods of $m = 5$ minutes, I_p is a range of minutes for each period:

```
m, n = 5, 3 # 5 minutes per period, 3 periods
I = [((p - 1) * m + 1):(p * m) for p in 1:n]
```

```
3-element Vector{UnitRange{Int64}}:
 1:5
```

```
6:10
11:15
```

Parameters?

Question: What could be possible parameters?

- $q_{o,d,p}$ - Queued passengers from station o to d with $o, d \in \mathcal{O}$ in p
- τ_e - Travel time (min) of the arcs $e \in \mathcal{E}$
- c_e - Max. allowed arc entry rate e per minute with $e \in \mathcal{E}$
- c_o^{min} - Min. station entry rate o per minute with $o \in \mathcal{O}$
- c_o^{max} - Max. station entry rate o per minute with $o \in \mathcal{O}$
- α - Safety factor applied to all capacities $(0 < \alpha < 1)$

Metro Movement

Question: How do people move inside the metro?

- Simple network: assume people use the **shortest path**
- Use **Dijkstra's algorithm** to compute it
- From each station to all other stations

> **Tip: We can save the results in a new set $\mathcal{C}_{o,d}$**
> This will help us to model the movement inside the metro.

Shortest Paths (SP)

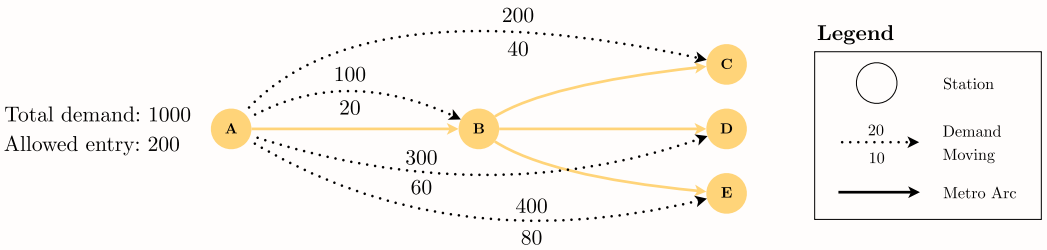
- $\mathcal{C}_{o,d}$ - Set of arcs $e \in \mathcal{E}$ on the SP from o to d with $o, d \in \mathcal{O}$

> **Note**
> The arc travel times τ_e are only the **input** to Dijkstra's algorithm. From the resulting paths we compute the travel time parameter the model actually uses:

- $\tau_{o,e}$ - Travel time (min) on the SP from station $o \in \mathcal{O}$ to arc $e \in \mathcal{E}$

People Spreading

Question: How do people spread?



Ratio of Origin-Destination

- We **cannot control** the destination of the passengers
- Thus we assume that people spread **based on destination**

$$\frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \quad \forall o, d \in \mathcal{O}, p \in \mathcal{P}$$

> **Note**
> Based on the ratio of the different destinations d to the total queue for each station $o \in \mathcal{O}$ in each period $p \in \mathcal{P}$. Note that the summation needs its **own index** f over all destinations, as d is already taken by the ratio's destination.

Variables & Objective

Decision Variable?

> **Our goal is to:**
> Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

> **Tip**
> There is **one queue for all passenger flow directions**, as managing multiple queues would be too complex for the planners!

Decision Variable

> **We need the following sets:**
> - All the metro stations, $o \in \mathcal{O}$
> - All periods under observation, $p \in \mathcal{P}$

Question: What could be our decision variable?

- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

Objective Function?

> **Our main objective is to:**
> Regulate the **inflow** at **each individual station** to **minimize the queues** outside the metro stations based on the allowed inflow while **adhering to the capacity constraints.**

Question: How again are queues minimized?

- By the allowed inflow $X_{o,p}$ subtracted from the queue

Objective Function

> **We need the following parameters and variables:**
> - $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in period p
> - $X_{o,p}$ - Allowed inflow (per minute) at station o in period p

Question: What could be our objective function?

$$\text{Minimize} \quad \sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} \left(\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \right)$$

Objective Function Revisited

Question: What does the solver really maximize here?

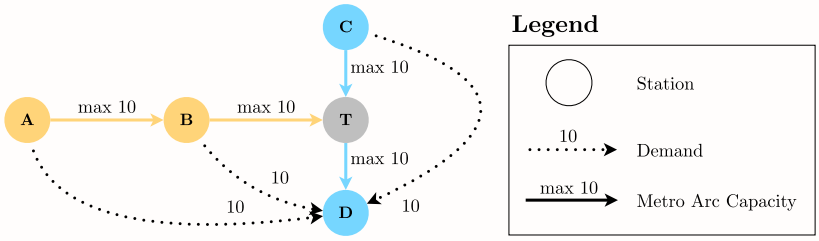
- The queued passengers $q_{o,d,p}$ are **parameters**, not variables
- Their sum is a **constant**: the solver cannot change it

> **Note: Minimizing queues = maximizing admissions**
> Minimizing the objective is exactly the same as **maximizing the total admitted passengers** $\sum_{o \in \mathcal{O}} \sum_{p \in \mathcal{P}} m \times X_{o,p}$. This is also why the model gets away without explicit queue dynamics.

Constraints

Necessary Constraints

Question: What constraints do we need?



- The capacity of each arc is **not exceeded**
- Do not dispatch more people than are queued
- Do not dispatch less than the minimum allowed inflow

Things are going to

get a little

complicated now!

Central Question

Question: When do people flowing into the metro system change the arcs?

- People enter metro station at o with destination d
- They will lead to a usage of an arc e on their SP
- This usage depends on their path and the travel times

> **Tip: You'll likely know what we need now!**
> We can add a new set $\mathcal{R}_{e,t}$ to help us with this.

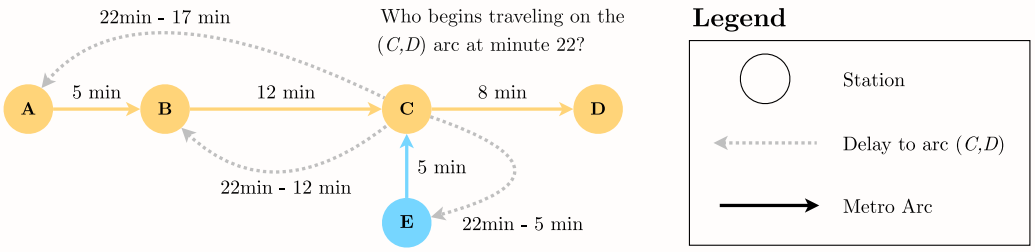
Set of Time-Delays

$$\mathcal{R}_{e,t} = \{(o,d,p) \,|o,d \in \mathcal{O}, p \in \mathcal{P}, q_{o,d,p} > 0, \\ e \in \mathcal{C}_{o,d}, t - \tau_{o,e} \in I_p\} \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

Question: Who can explain this set?

The set $\mathcal{R}_{e,t}$ contains **all combinations** (o,d,p) which trigger a **capacity utilization** of arc e in minute t.

Small Example



> **Tip**
> The set contains all o-d pairs and periods that result in passengers entering arc (C,D) at minute 22. With $m = 2$, it is $\{(A,D,3),(B,D,5),(C,D,11),(E,D,9)\}$.
>
> Check it yourself, e.g. from A: $\tau_{A,(C,D)} = 5 + 12 = 17$ and $22 - 17 = 5 \in I_3 = \{5,6\}$.

Essentially, we can use this set to compute the utilization of an arc at each minute based on the inflow level at the different stations and periods.

Ensure Capacity Utilization?

! **The goal of this constraint is to:**
Ensure that the capacity of each arc is not exceeded at any minute.

i **We need the following variable, parameter and set:**
- $q_{o,d,p}$ - people queued to travel from station $o \in \mathcal{O}$ to station $d \in \mathcal{O}$ in p
- c_e - people max. allowed to enter arc e per minute with $e \in \mathcal{E}$
- $\mathcal{R}_{e,t}$ - mapping of station entries to arc e in minute t with $e \in \mathcal{E}$ and $t \in \mathcal{T}$
- α - safety factor applied to all capacities $(0 < \alpha < 1)$
- $X_{o,p}$ - the allowed inflow per minute at metro station o in the period p

Ensure Capacity Utilization

Question: What could be the constraint?

$$\sum_{(o,d,p) \in \mathcal{R}_{e,t}} X_{o,p} \times \frac{q_{o,d,p}}{\sum_{f \in \mathcal{O}} q_{o,f,p}} \leq \alpha \times c_e \quad \forall e \in \mathcal{E}, t \in \mathcal{T}$$

i **Note**
Here, we combine the inflow at each station based on the o-d proportion and let the people spread through the network, checking that no arc is over-utilized at any minute.

Bound the Inflow Rate?

! **The goal of this constraint is to:**
Ensure that the inflow rate at each station stays within operational bounds.

i **We need the following variables and parameters:**
- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- c_o^{min} - Min. station entry rate o per minute
- c_o^{max} - Max. station entry rate o per minute
- α - Safety factor applied to all capacities $(0 < \alpha < 1)$

Bound the Inflow Rate

Question: What could be the constraint?

$$c_o^{min} \leq X_{o,p} \leq \alpha \times c_o^{max} \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

i **Note**
This constraint bounds the inflow rate at each station between the **minimum and maximum** allowed entry rates, ensuring compliance with **operational capacity limits**.

Dispatch Only Available People?

! **The goal of this constraint is to:**
Ensure that we do not dispatch more people than are queued.

i **We need the following variables and parameters:**
- $X_{o,p}$ - Allowed inflow (per minute) at station o in period p
- $q_{o,d,p}$ - People queued to travel from station o to d with $o, d \in \mathcal{O}$ in p
- m - period length in minutes

Dispatch Only Available People

Question: What could be the constraint?

$$\sum_{d \in \mathcal{O}} q_{o,d,p} - m \times X_{o,p} \geq 0 \quad \forall o \in \mathcal{O}, p \in \mathcal{P}$$

i **Note**
This constraint ensures **non-negative remaining queues**, preventing the solver from dispatching more passengers than available in the queue.

A Hidden Pitfall

Question: What if a queue is smaller than $m \times c_o^{min}$?

- The lower bound forces $m \times X_{o,p} \geq m \times c_o^{min}$
- But we may only dispatch up to $\sum_{d \in \mathcal{O}} q_{o,d,p}$ people
- Example: $c_o^{min} = 2$, $m = 15$, but only 20 people queued
- A quiet station late at night makes the model **infeasible**!

Question: How could we fix this?

- Set $c_o^{min} = 0$, cap the bound at the queue, or soften the constraint

Metro Inflow Model

subject to:

Model Characteristics

Characteristics

Questions: On model characteristics

- Is the model formulation linear/ non-linear?
- What kind of variable domains do we have?

> **Note: The answers**
> It is a **linear program**, as objective and constraints are linear. The variable $X_{o,p} \geq 0$ is **continuous**: fractional inflow rates per minute are perfectly acceptable here.

Model Assumptions

Questions: On model assumptions

- What assumptions have we made?
- What are likely issues that can arise if applied?
- Have we thought in detail about queues?
- Are shortest paths a feasible assumption?

In Practice

Metro Inflow Optimization

Question: Can this be applied?

Metro Inflow Problem

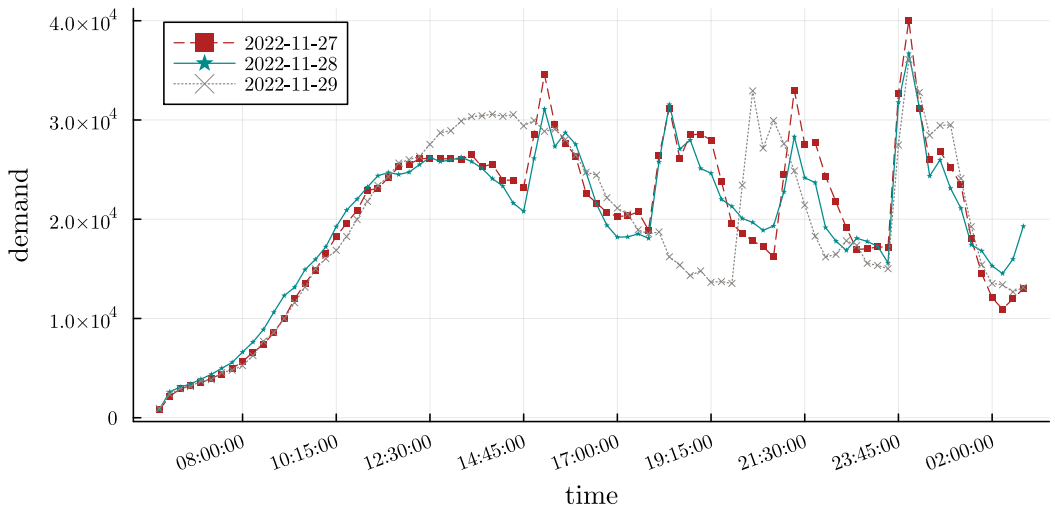
- Solved very fast **within seconds** for realistic problem sizes
- **But** we cannot plan or control the metro inflow
- Queues are **too simplified** with passengers disappearing

Question: Any ideas, how the current model could be improved or how it could be embedded into a heuristic?

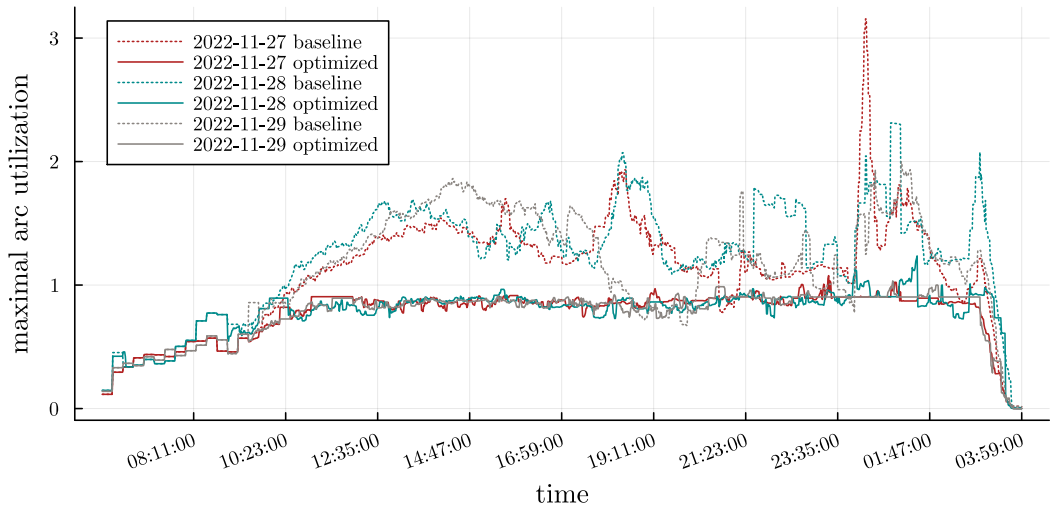
Heuristic: Step-Wise

- Solve the model for the time-horizon of a **few periods**
- Fix the inflow in the **current first period**
- Decrease capacity in the network **based on the inflow**
- Transfer **remaining queues** into the subsequent period
- **Solve the model again**
- **Repeat**, until the inflow is computed for all periods

Transport Demand



Utilization Analysis



Key Results

- Model reduces capacity violations from **25,941 to just 4**
- Maximum arc utilization drops from **316% to 109%**
- **81% of passengers** accommodated safely
- Computation time: **~1.4 seconds** per planning period

> **Note: Safety vs. Throughput Tradeoff**
> The remaining 19% of passengers experience external queuing - this is the price for preventing dangerous overcrowding inside the system.

Didn't We Bound the Utilization?

Question: Our constraint enforces $\alpha \times c_e$ with $\alpha < 1$. How can utilization still reach 109%?

- The constraint binds the model for the **forecast** demand
- The 109% occurs in the **rolling-horizon simulation** with realized demand
- Realized flows can deviate from the forecast, but far less than **316%**

An Ordinary Day in Shanghai

Question: Does inflow control also work without a World Cup?

Scalability: Shanghai Metro

- Validated on **302 stations** (~8× more than Doha's 37)
- **99% reduction** in capacity violations
- Computation: **< 2 minutes** per period
- Works for routine **peak-hour demand**

> **Tip: Generalizable Approach**
> The method scales from event-driven demand (Qatar World Cup) to routine metropolitan operations (Shanghai rush hour).

Implementation

- **Assumption of known destinations** is strong
- Movements **seemed to follow our forecasts**
- We did achieve our goal of **metro inflow control**
- Simulation was used to **estimate the inflows**

> **Note**
> Few dangerous situations, especially at the FIFA Fan Fest, were **handled well** by the authorities.

Comparison with Others

Feature	Other Methods	Our Approach
FIFO Enforcement	Rarely	**Yes**
Single Queue	Often separate	**Yes**
Real-Time Capable	Some	**Yes (seconds to minutes)**
Network Scale	Up to 327 stations	Comparable (302 stations)

> **Note: State-of-the-Art**
> First network-level approach combining FIFO enforcement, single-queue modeling, and real-time computational performance.

Wrap Up

- Model can help to achieve a **good balance**
- Can be adapted easily to **any metro system worldwide**
- Especially interesting for **larger Asian cities**

> **And that's it for today's lecture!**
> We now have covered a metro inflow control problem based on a real-world application and are ready to start solving some new tasks in the upcoming tutorial.

Questions?

Literature

For more interesting literature to learn more about Julia, take a look at the literature list of this course.